Code No: 181AP **R22**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech I Year I Semester Examinations, March/April - 2023

PROGRAMMING FOR PROBLEM SOLVING

(Common to CSE, IT, CSIT, CE(SE), CSE(CS), CSE(DS), CSD)

Time: 3 Hours **Max. Marks: 60**

Note: This question paper contains two parts A and B.
i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

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PART- A

(10 Marks)

1.a)	Give example for an Identifier.	[1]
b)	What is syntax error?	[1]
c)	What does strstr() function do?	[1]
d)	Define Pointer.	[1]
e)	What fseek() function do?	[1]
f)	What is define preprocessor command?	[1]
g)	What is void return type indicate?	[1]
h)	Define function signature.	[1]
i)	Write Time complexity of Binary search.	[1]
j)	Write Time complexity of Insertion sort.	[1]

ART - B

(50 Marks)

2.a)	Write pseudo code to find biggest number of a given set.	
b)	Write the syntaxes of While and Do-While loops. Explain. Differentiate between them.	[4+6]

OR

3.a)	Write an algorithm to find the roots of a Quadratic equation.	
b)	List out Bitwise operators. Explain each one along with their precedence and associativity.	[4+6]
4.a)	Define an Array. Write a C code for Matrix multiplication using arrays.	
b)	Write about Pointer Arithmetic Operations.	[6+4]

OR

5.a)	Define Union? Write the advantages of Union over Structure? Justify your answer?	
b)	Write a C code to find string length?	[4+6]

6.a) Write about various operations that can be performed on files. Write various modes of file operations.

b) Write a C code to read and print the contents of a file. [5+5]

OR

7.a) What is the need of Pre-processor commands? Explain.

b) Write a C code to find the number of characters in a file. [4+6]

8.a) What is dynamic memory allocation? Why it is required explain?

b) Write a C code to find Factorial of a given number using Recursion. [5+5]

OR

9.a) Differentiate call by value and call by reference.

b) Write a C code to print Fibonacci sequence using recursion. [4+6]

10.a) Write Linear search algorithm? Explain with example.

b) Write two applications of searching. [8+2]

OR

11.a) What is sorting? Explain Selection sort algorithm with example.

b) What is Time complexity? [8+2]

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